

Assignment 2: Building a Todo List Application with React Native

Objective

- Build a simple React Native application to manage a todo list.
- Utilize core React Native components such as TextInput, Button, and FlatList.
- Manage application state using database(sqlite, firebase, monogoDB,...).
- Implement screen navigation using React Navigation.
- **Extension:** Add a "Remember Password" feature to automatically log in on subsequent app launches.

Functional Requirements

1. Login Screen:

- Display two input fields: Username and Password.
- Include a "Login" button to handle login.
- **Add a "Remember Password" checkbox:**
 - If checked, save the login credentials (username and password) to automatically log in on future app launches.
 - If unchecked, do not save credentials, requiring the user to log in again when the app is opened.
- If login is successful (simulated check, e.g., non-empty username and password), navigate to the main screen.
- If login fails, display an error message (alert).
- **Auto-login:**
 - On app launch, check for saved credentials. If valid credentials exist, navigate directly to the main screen without showing the login screen.

2. Main Screen:

- Display a list of current tasks using FlatList.
- Include an input field (TextInput) and an "Add" button to add new tasks.
- Each task in the list should display:
 - **Task Title:** Text describing the task.
 - **"Complete" Button:** Toggle the task's completion status.
 - **"Delete" Button:** Remove the task from the list.
- When the "Complete" button is pressed:

- The task title is displayed with a strikethrough (textDecorationLine: 'line-through') or styled differently (e.g., change text color) to indicate completion.
- When the "Delete" button is pressed:
 - The task is removed from the list.
- Data does not need to be persisted (task state is lost when the app is reloaded).

3. **State Management:**

- Use database(sqlite, firebase, monogoDB,...) to manage the state of the task list (add, delete, toggle completion).
- The state includes a list of tasks, each with the following properties:
 - id: Unique identifier (e.g., timestamp or simple UUID).
 - title: Task title.
 - completed: Completion status (boolean).
- **Store Login Credentials:**
 - Use AsyncStorage to save and retrieve login credentials (username and password) when the "Remember Password" option is selected.